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ii. **Active transport:** It is a biological process and it runs based on the energy obtained from respiration. It is an uphill process.

11.2 Cell to Cell Transport

Cell to cell or short distance transport covers the limited area and consists of few cells. They are the facilitators or tributaries to the long-distance transport. The driving force for the cell to cell transport can be passive or active (Figure 11.1). The following chart illustrate the various types of cell to cell transport:

Diffusion: The net movement of molecules from a region of their higher concentration to a region of their lower concentration along a concentration gradient until an equilibrium is attained.

Characteristics of diffusion

i. It is a passive process, hence no energy expenditure involved.
ii. It is independent of the living system.
iii. Diffusion is obvious in gases and liquids.

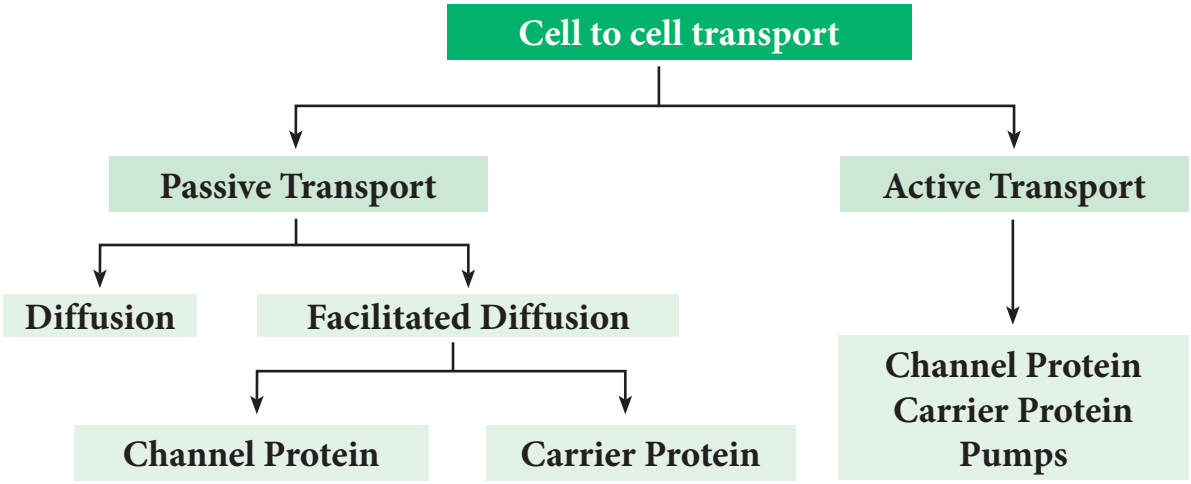


Figure 11.1: Cell to cell transport

11.2.1 Passive Transport

1. Diffusion

When we expose a lightened incense stick or mosquito coil or open a perfume bottle in a closed room, we can smell the odour everywhere in the room. This is due to the even distribution of perfume molecules throughout the room. This process is called **diffusion**.

In **diffusion**, the movement of molecules is continuous and random in order in all directions (Figure 11.2).

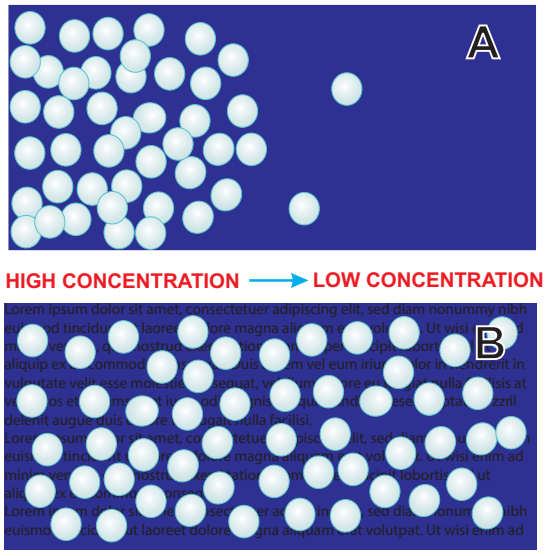


Figure 11.2: Distribution of molecules in diffusion (A) Initial stage (B) Final stage